

Susan Sapkota

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Summary

I am a Computer Engineering student at IOE Purwanchal Campus with a strong interest in Artificial Intelligence and Machine Learning. I have a background in programming, problem solving, and developing practical software projects. I have hands-on experience with Python, machine learning, deep learning, NLP, computer vision, OCR, and RAG-based applications. I enjoy learning new technologies and applying them to real-world problems. My goal is to further develop my AI/ML knowledge and technical skills through practical experience and research-oriented projects.

Technical Skills

Programming: Python, C++, C, JavaScript

Machine Learning: scikit-learn, NumPy, Pandas, Matplotlib

Deep Learning & AI: PyTorch, TensorFlow, Transformers, NLP, Computer Vision, OCR

Generative AI: RAG, LLMs, Embeddings

Web Development: React.js, Next.js, Node.js, Express.js, REST APIs

Databases: MongoDB, MySQL

Tools: Git, GitHub, Linux, VS Code, Postman, DVC, MLflow

Projects

Kidney Disease Classification | *TensorFlow, OpenCV, scikit-learn, DVC, MLflow*

- Developed a deep learning pipeline for kidney disease image classification using TensorFlow and computer vision techniques.
- Built a modular training workflow with configuration management, DVC-based pipeline reproduction, and MLflow experiment tracking.
- Integrated the trained model with a Flask-based application for inference.

Retrieval-Augmented Generation (RAG) System | *LangChain, FAISS, ChromaDB, Sentence Transformers*

- Built a document-based RAG system to retrieve relevant information and generate responses from user queries.
- Configured document processing, embeddings, vector storage, and similarity-based retrieval using FAISS and ChromaDB.
- Integrated LangChain and Sentence Transformers to develop the retrieval and generation pipeline.

Movie Recommendation System | *Python, Machine Learning, Streamlit*

- Built a machine-learning-based movie recommendation system using Python and Streamlit.
- Created recommendation logic to generate relevant movie suggestions from movie data.
- Developed an interactive Streamlit interface for users to explore recommendations.

Certifications

- **Python Programming (Basic to Intermediate)** — Programiz
- **JavaScript Programming (Basic to Intermediate)** — Programiz

Achievements

- **LeetCode:** Solved 216+ Data Structures and Algorithms problems and completed a 100-day problem-solving streak.

Education

Bachelor of Engineering in Computer Engineering

IOE Purwanchal Campus, Dharan

3rd Year

Expected Graduation: 2027

Higher Secondary Education (+2)

Shree Secondary School, Imiliya

GPA: 3.40